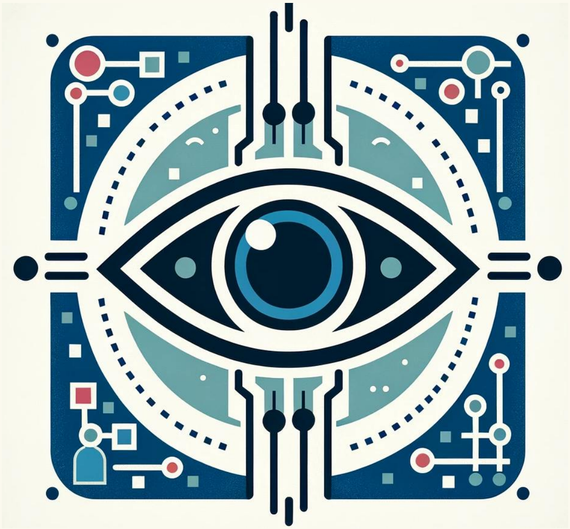


COMPUTER VISION

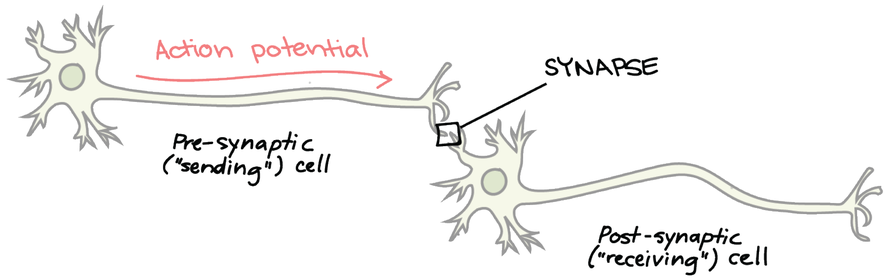
Neural Networks

(extra slides)

Luis F. Teixeira | M.EIC+M.IA | FEUP

Artificial Neural Networks

- Idea: mimic the brain to do computation

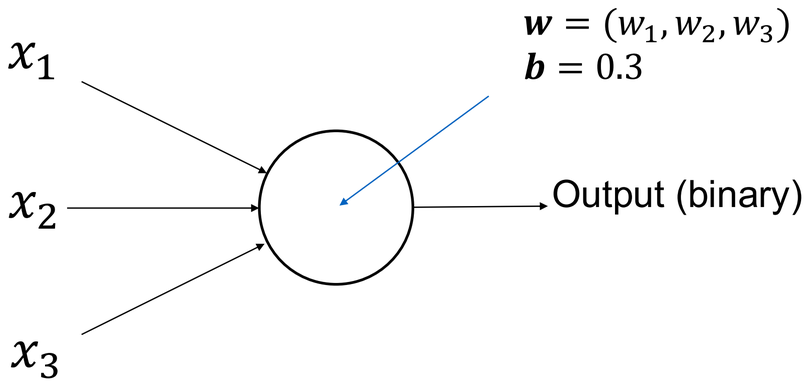


Source: Khan Academy

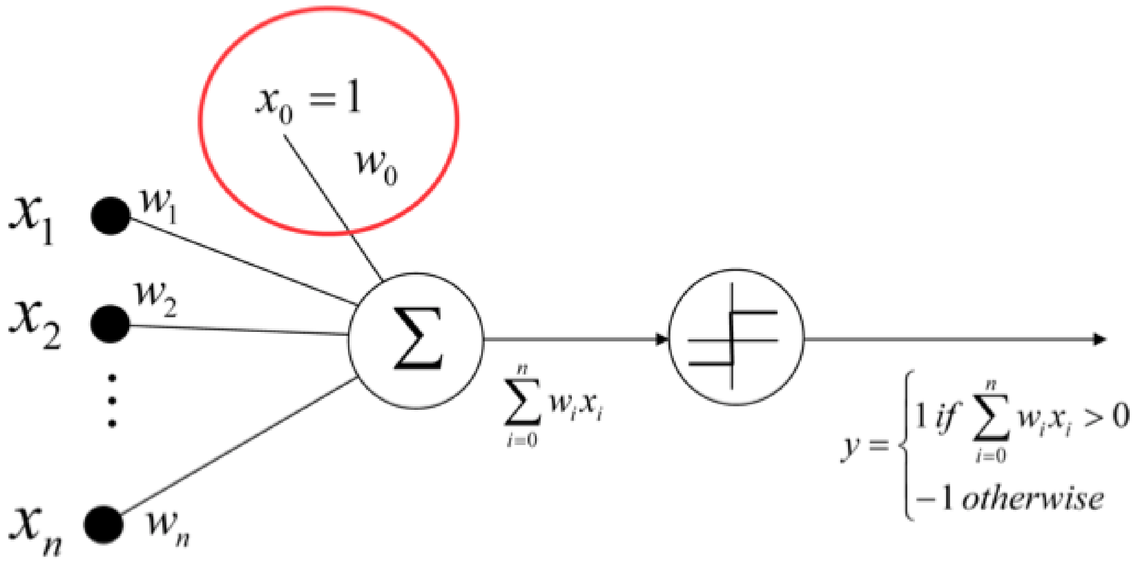
- Artificial neural network:
  - Nodes (a.k.a. units) correspond to neurons
  - Links correspond to synapses
- Computation:
  - Numerical signal transmitted between nodes corresponds to chemical signals between neurons
  - Nodes modifying numerical signal corresponds to neurons firing rate

Perceptron

- Basic building block for composition is a *perceptron* (Rosenblatt 1958)
- Vector of weights *w* and a ‘bias’ *b*



Perceptron – Decision

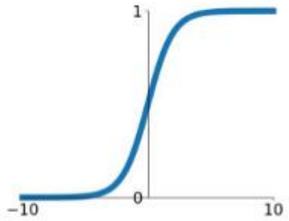


$$y = sign(w^T x)$$

Activation functions

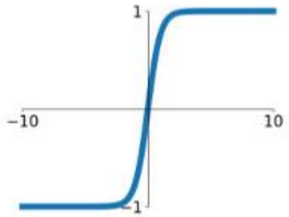
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



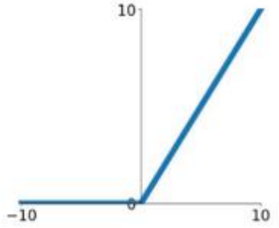
tanh

$$\tanh(x)$$



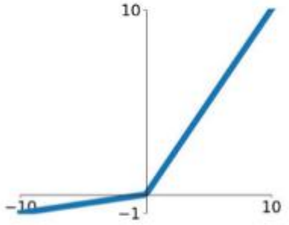
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$



Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$

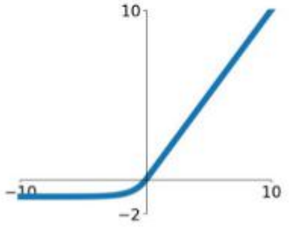
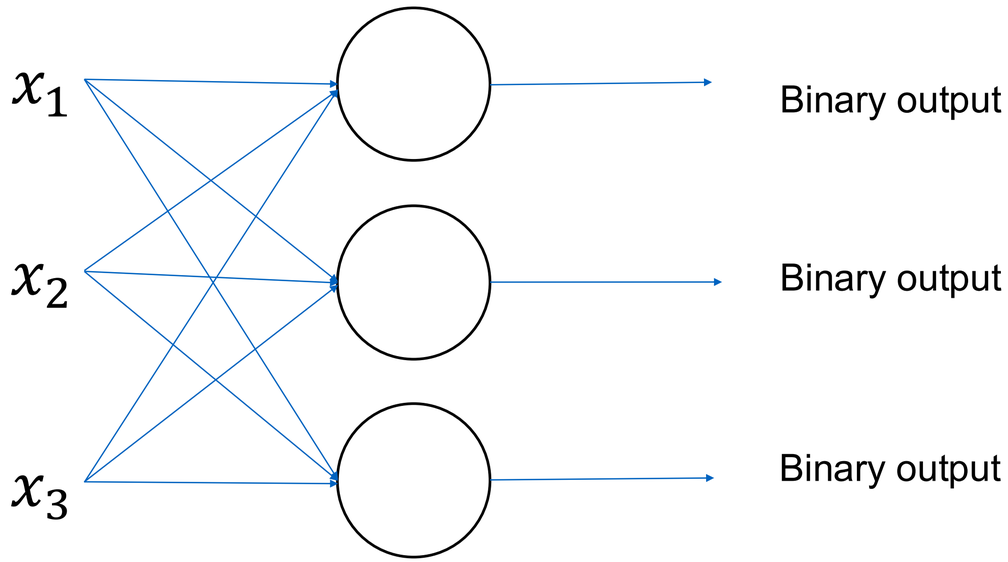


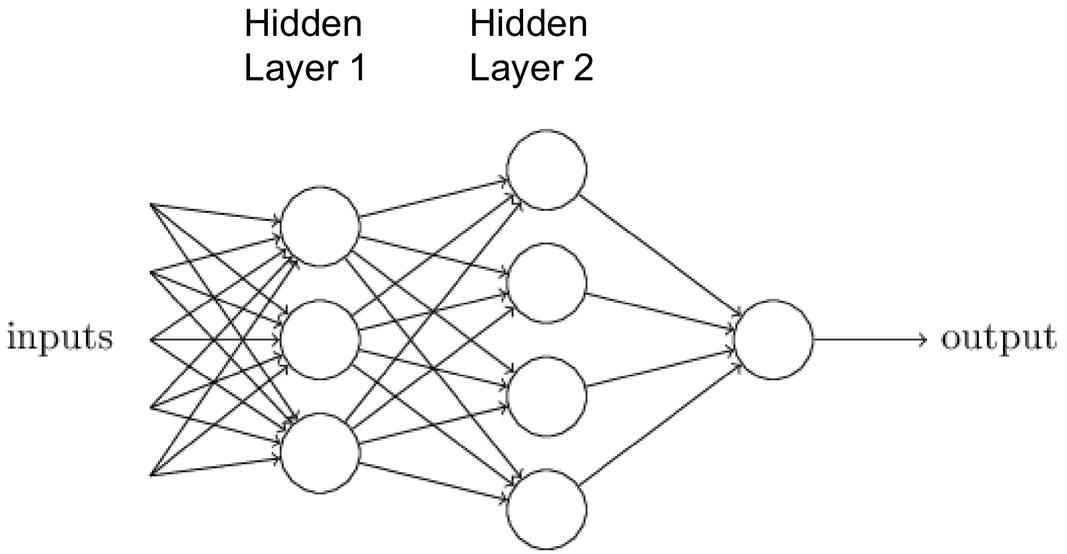
Image source: https://medium.com/@shrutijadon10104776/survey-on-activation-functions-for-deep-learning-9689331ba092

Neural Networks - multiclass

- Add more perceptrons



Multi-layer perceptron (MLP)



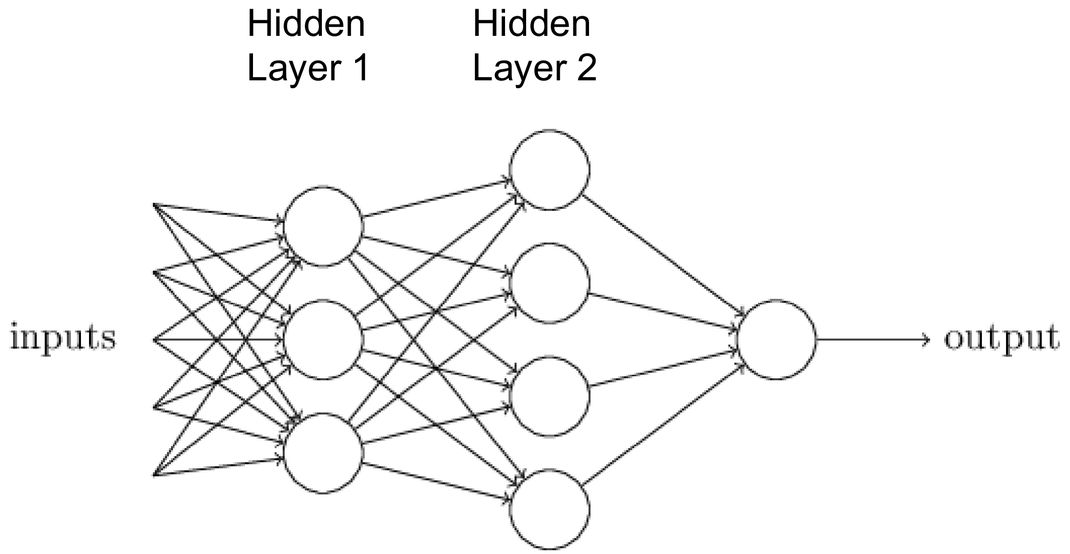
Sets of layers and the connections (weights) between them define the **network architecture**.

Each layer receives its inputs from the previous layer and forwards its outputs to the next layer

Explanation in 3Blue1Brown about the multi-layer perceptron
https://www.youtube.com/watch?v=aircAruvnKk

adapted from Nielsen

Multi-layer perceptron (MLP)

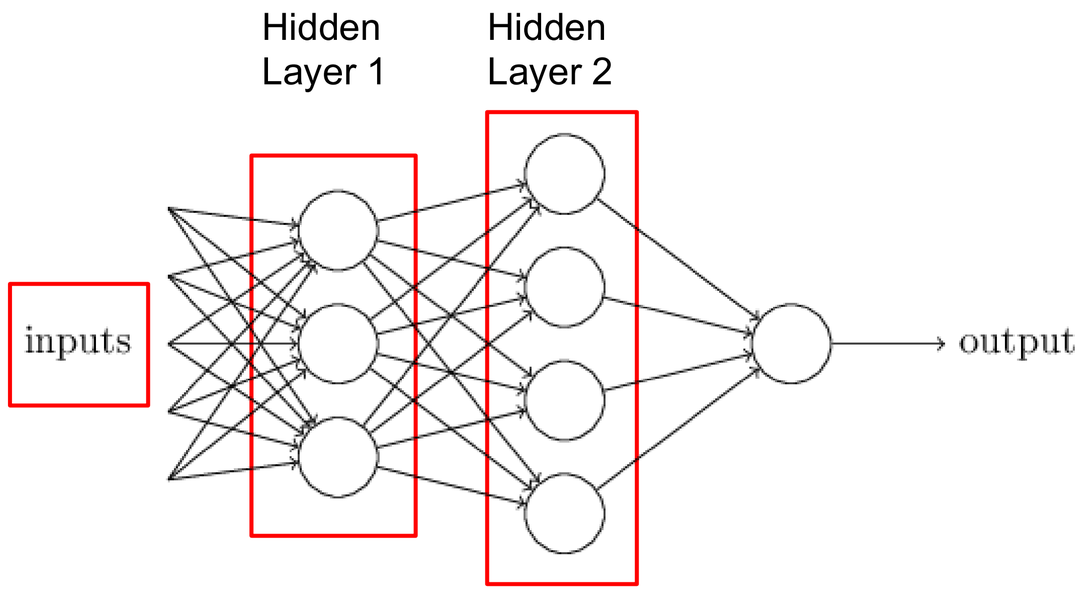


To handle more **complex problems** (than linearly separable ones) we need multiple layers → combination of linear boundaries which allow the **separation of complex data**

Theoretically, using at least two layers (one hidden + one output), with enough hidden units, **we can approximate any function**

adapted from Nielsen

Multi-layer perceptron (MLP)

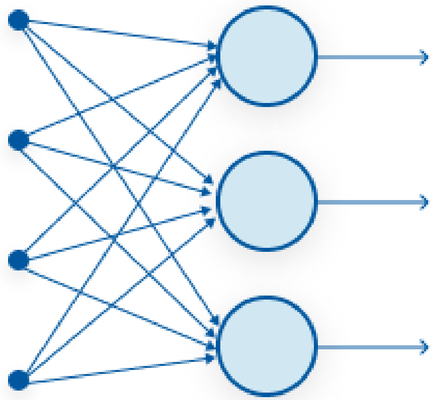


It's all just matrix multiplication!
GPUs -> special hardware for fast/large matrix multiplication.

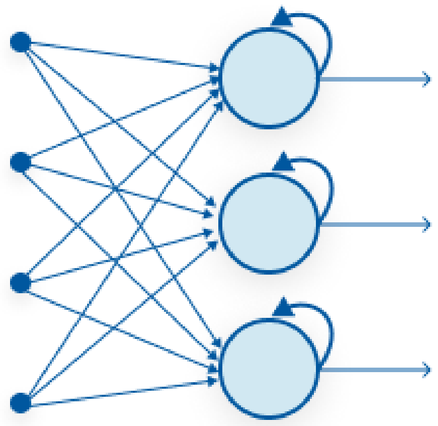
adapted from Nielsen

Network structure

- Feed-forward network
  - Directed **acyclic** graph
  - No internal state
  - Simply computes outputs from inputs



- Recurrent network
  - Directed **cyclic** graph
  - Dynamical system with internal states
  - Can memorize information



Adapted from Pascal Poupart

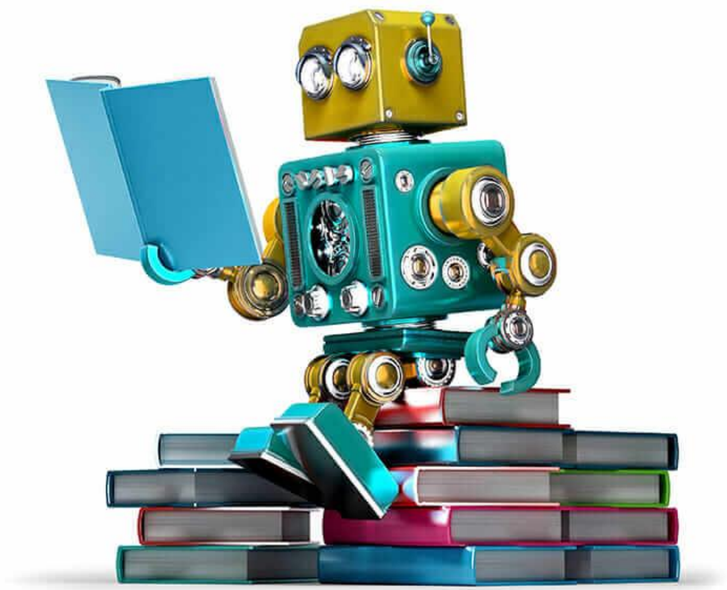
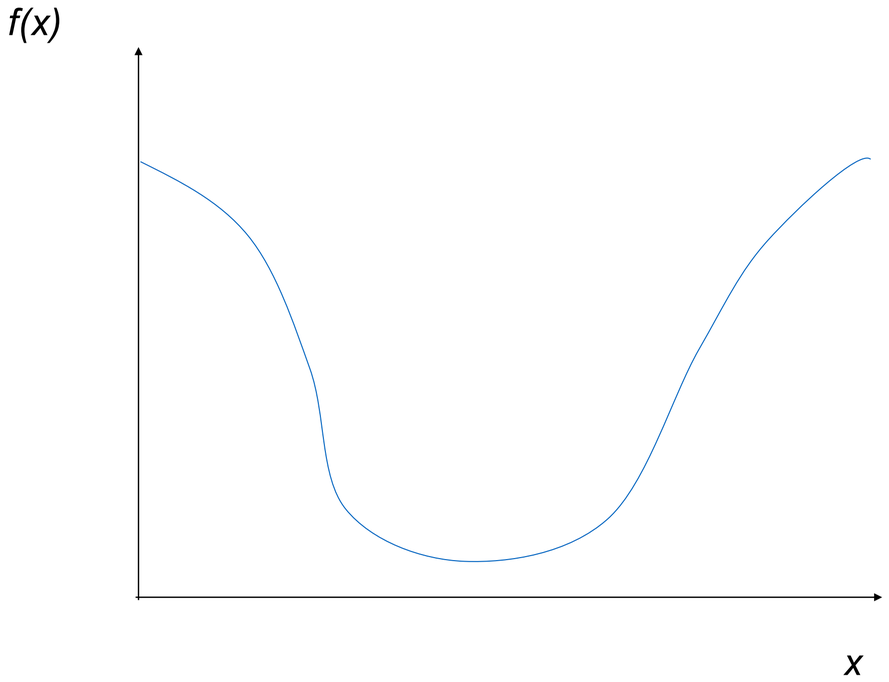


Image: https://redshift.autodesk.com/machine-learning/

Next step? Training a Neural Network

- Learning the weight matrices ***W*** → optimization problem
- Most common algorithm: gradient descent

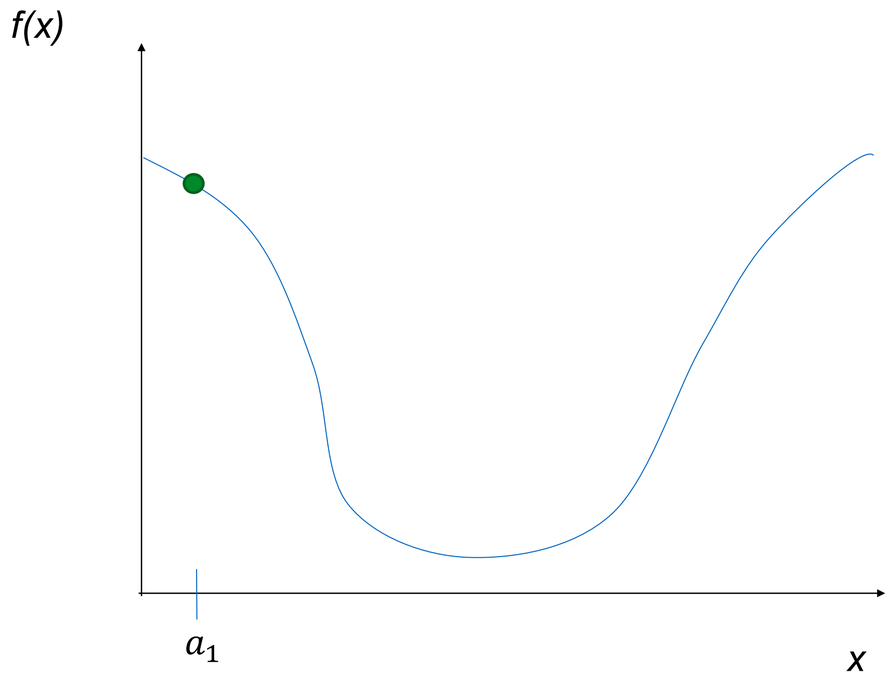
General approach



adapted from James Tompkin

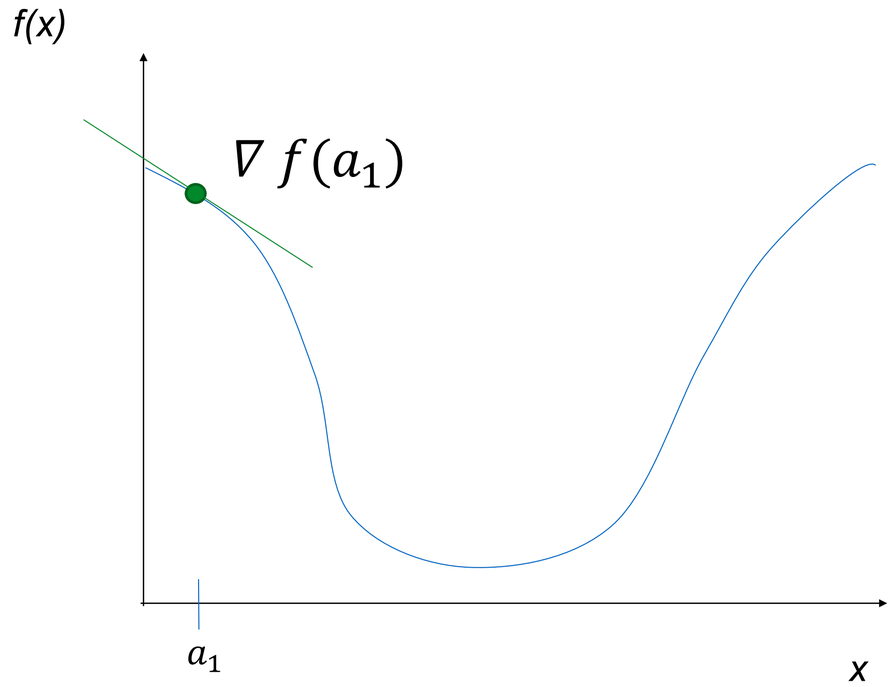
General approach

Pick random starting point.



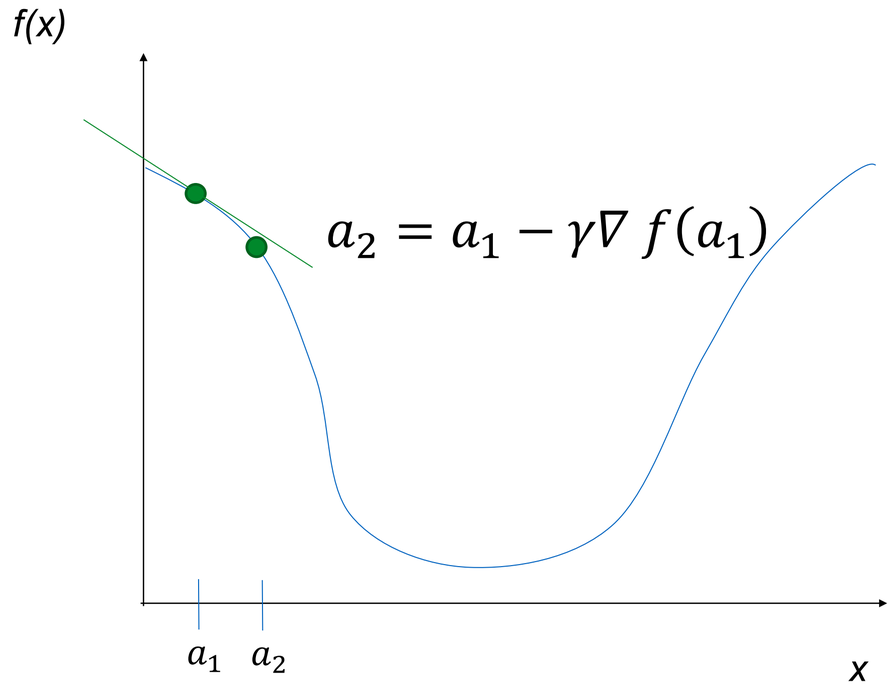
General approach

Compute gradient at point (analytically or by finite differences)



General approach

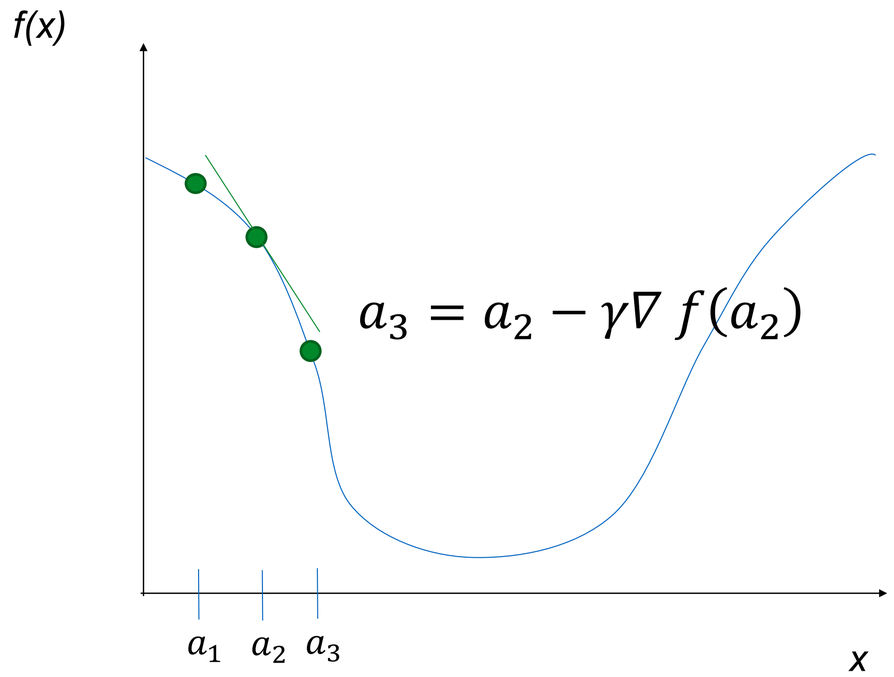
Move along parameter space in direction of negative gradient



γ = amount to move
= *learning rate*

General approach

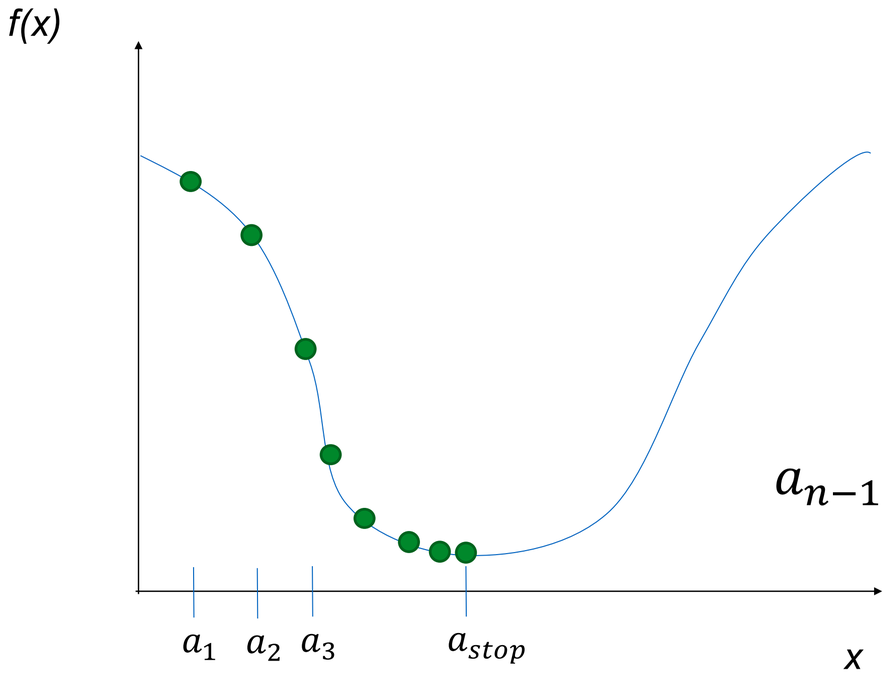
Move along parameter space in direction of negative gradient.



γ = amount to move
= *learning rate*

General approach

Stop when we don't move any more.



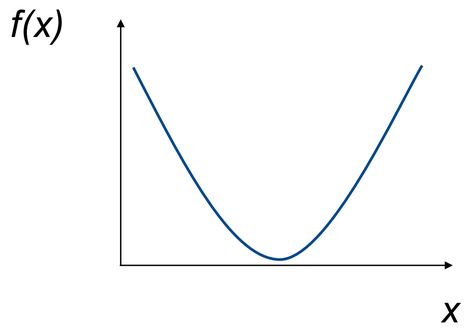
a_{stop}:

$$a_{n-1} - \gamma \nabla f(a_{n-1}) = 0$$

Gradient descent

- Optimizer for functions

- Guaranteed to find optimum for convex functions.
  - Non-convex = find *local* optimum.
  - Most vision problems aren't convex.



- Works for multi-variate functions.
  - Need to compute matrix of *partial derivatives ("Jacobian")*

Gradient descent

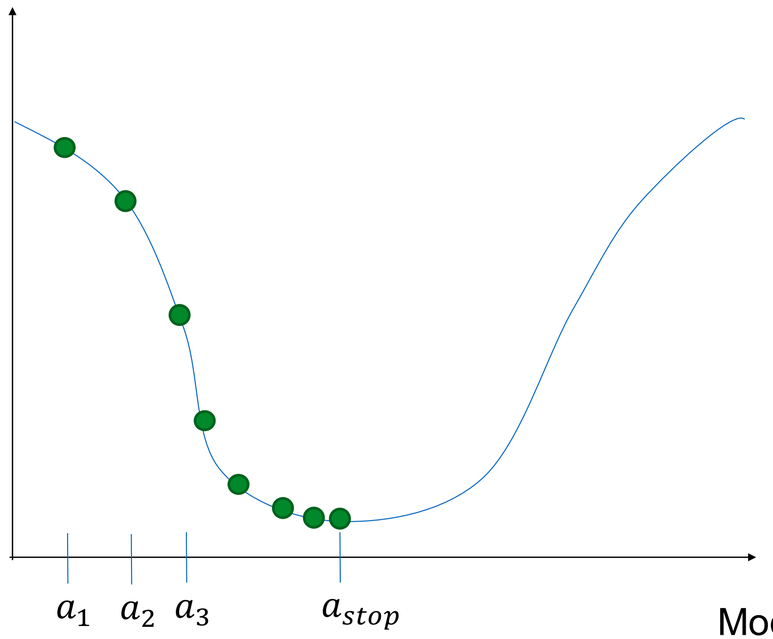
Train NN with gradient descent

- x^i, y^i = *n* training examples
- $f(\boldsymbol{x})$ = feed forward neural network
- L($\mathbf{x}$, y; θ) = some *loss function*

- *Loss function* (or cost function) measures how ‘good’ our network is at classifying the training examples wrt. the parameters of the model, i.e. the perceptron weights.

Train NN with gradient descent

Loss function
(Evaluate NN
on training data)



Model parameters
(perceptron weights)

adapted from James Tompkin

Sequential gradient descent

$$w_{ji} \leftarrow w_{ji} - \eta \frac{\partial L}{\partial w_{ji}}$$

Error or Loss

Learning rate or step length

- In practice, the training is typically done using **sequential gradient descent**, i.e. in each iteration (step), calculate the error and update the weights
- A complete pass over the training set is called an epoch
- How can we compute the gradient efficiently given an arbitrary network structure?

– Answer: backpropagation algorithm

  – propagating errors backward into the network

Sequential gradient descent

- If each iteration is done:
  - with every input, we have stochastic descent → **on-line training**
  - after accumulating error of all input samples → **batch learning**
  - as a mix of both → **mini-batch learning**
    - This is the usual setup, especially considering the hardware limitations when training models with large datasets

- Important: training with some samples at each step is not the same as with all the samples (batch) → error surface changes

Explanation in 3Blue1Brown about the training process
https://www.youtube.com/watch?v=IHZwWFHWa-w

What is an appropriate loss?

- Define some output threshold on detection
- Classification: compare training class to output class
- Zero-one loss L (per class)

$y = true\ label$
$\hat{y} = predicted\ label$

$$L(\hat{y}, y) = I(\hat{y} \neq y),$$

- Is it good?
  - Nope – it's a step function.
  - I need to compute the gradient of the loss.
  - This loss is not differentiable, and 'flips' easily.

Typical Loss functions

- Regression
  - Mean Squared Error (MSE) / L2
  - Mean Absolute Error (MAE) / L1
- Binary Classification
  - Binary Cross-Entropy (BCE)
  - Hinge Loss
- Multi-Class Classification
  - Multi-Class Cross-Entropy (CE)

$$\mathbf{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2$$

$$\mathbf{MAE} = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n}$$

$$\text{BCE} = -\frac{1}{m}\sum_{i=1}^{m}(y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i))$$

$$\text{CE} = -\frac{1}{m}\sum_{i=1}^{m} y_i \cdot \log(\hat{y}_i)$$

- The output of the last layer must be coupled with the loss function:
  - Regression → linear activation
  - Binary classification → sigmoid
  - Multiclass classification → softmax

Class probability estimation

Special function on last layer - ‘Softmax’:

- "squashes" a C-dimensional vector O of arbitrary real values to a C-dimensional vector σ(O) of real values in the range (0, 1) that add up to 1.
- Turns the output into a probability distribution on classes.

$$p(c_k = 1 | \boldsymbol{x}) = \frac{e^{o_k}}{\sum_{j=1}^{C} e^{o_j}}$$

For binary classification, with one output unit, the sigmoid activation function already outputs a probability distribution.

adapted from James Tompkin

Backpropagation algorithm

- Two phases:
  - Forward phase: compute output z_j, of each unit j

$$z_j = h(a_j) \; where \; a_j = \sum_i w_{ji} z_i$$

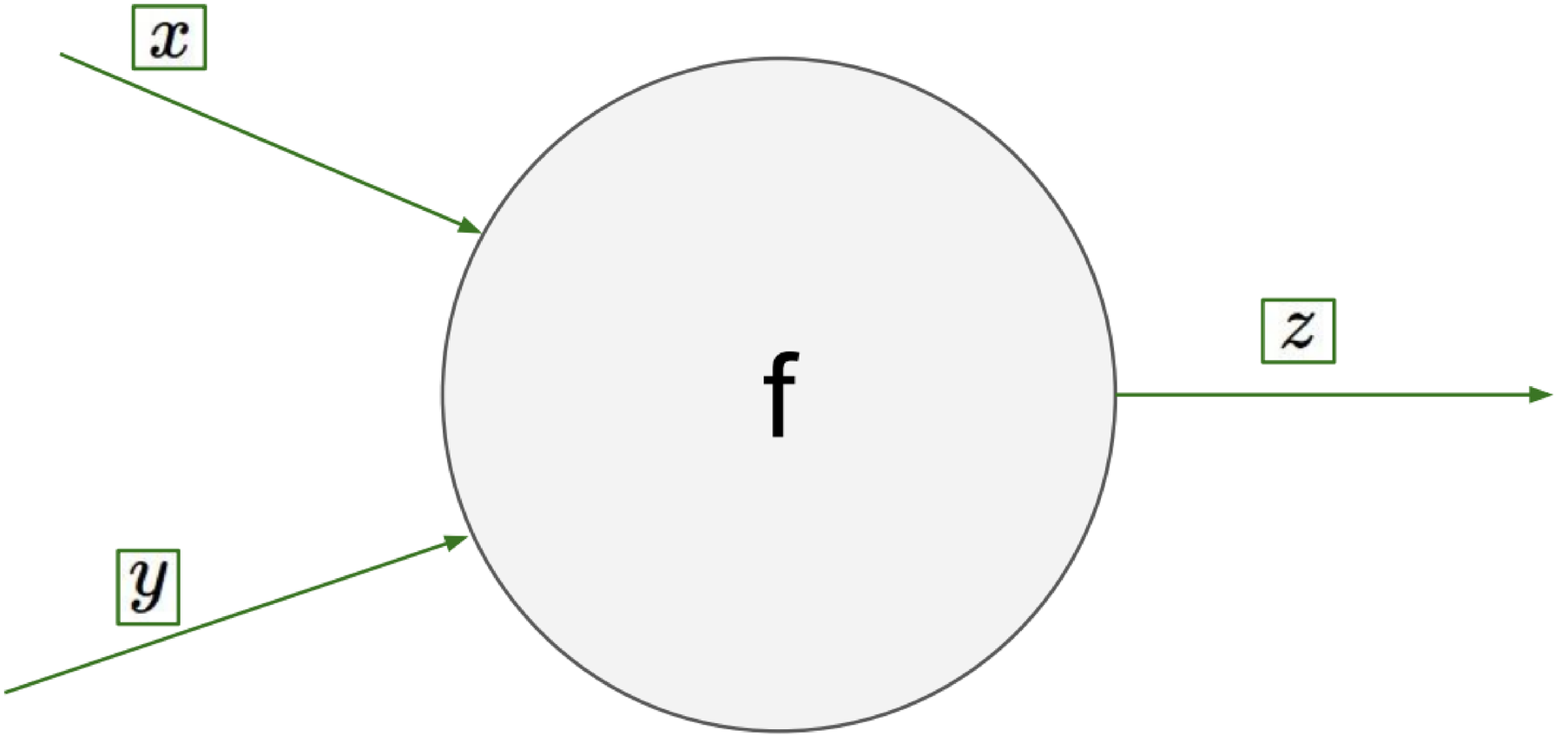
  - Backward phase: compute δ_j, of each unit j
    - Use chain rule to recursively compute gradient
      - For each weight w_{ji}: $\frac{\partial L}{\partial w_{ji}} = \frac{\partial L}{\partial a_j} \frac{\partial a_j}{\partial w_{ji}} = \delta_j \, z_i$

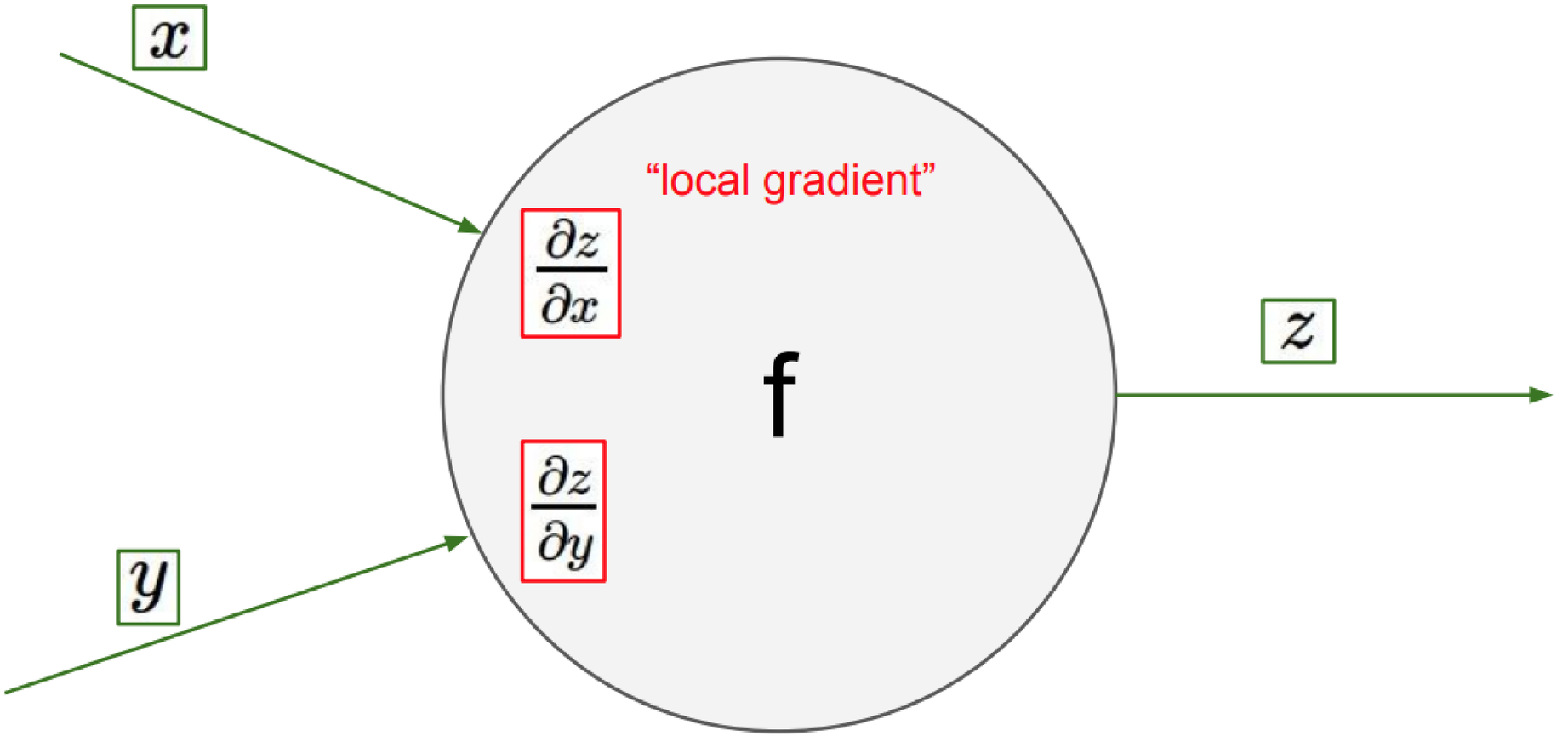
$$\delta_j = \begin{cases} h'(a_j)(z_j - y_j) \\ h'(a_j) \sum_k w_{kj} \delta_K \end{cases}$$

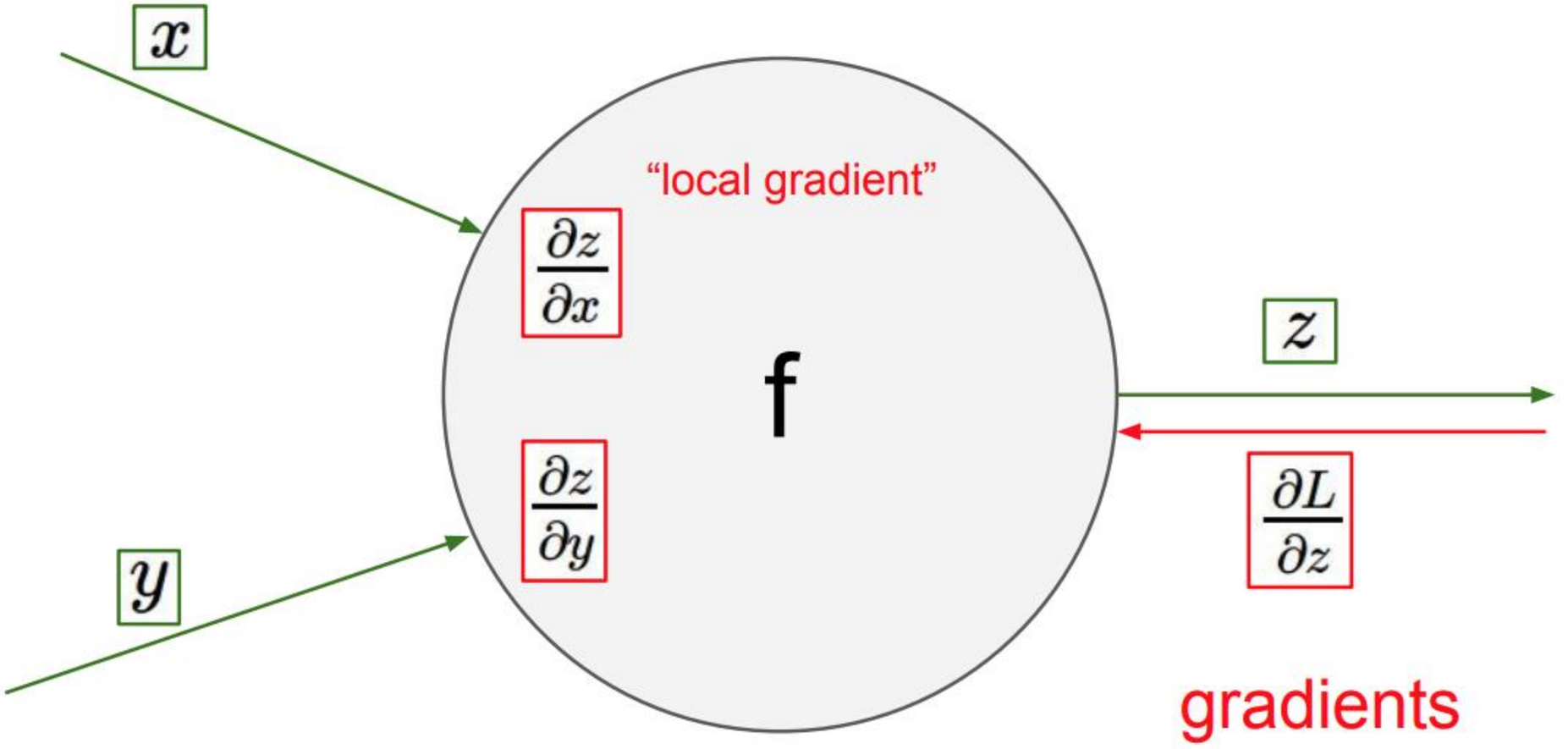
base case: j is an output unit

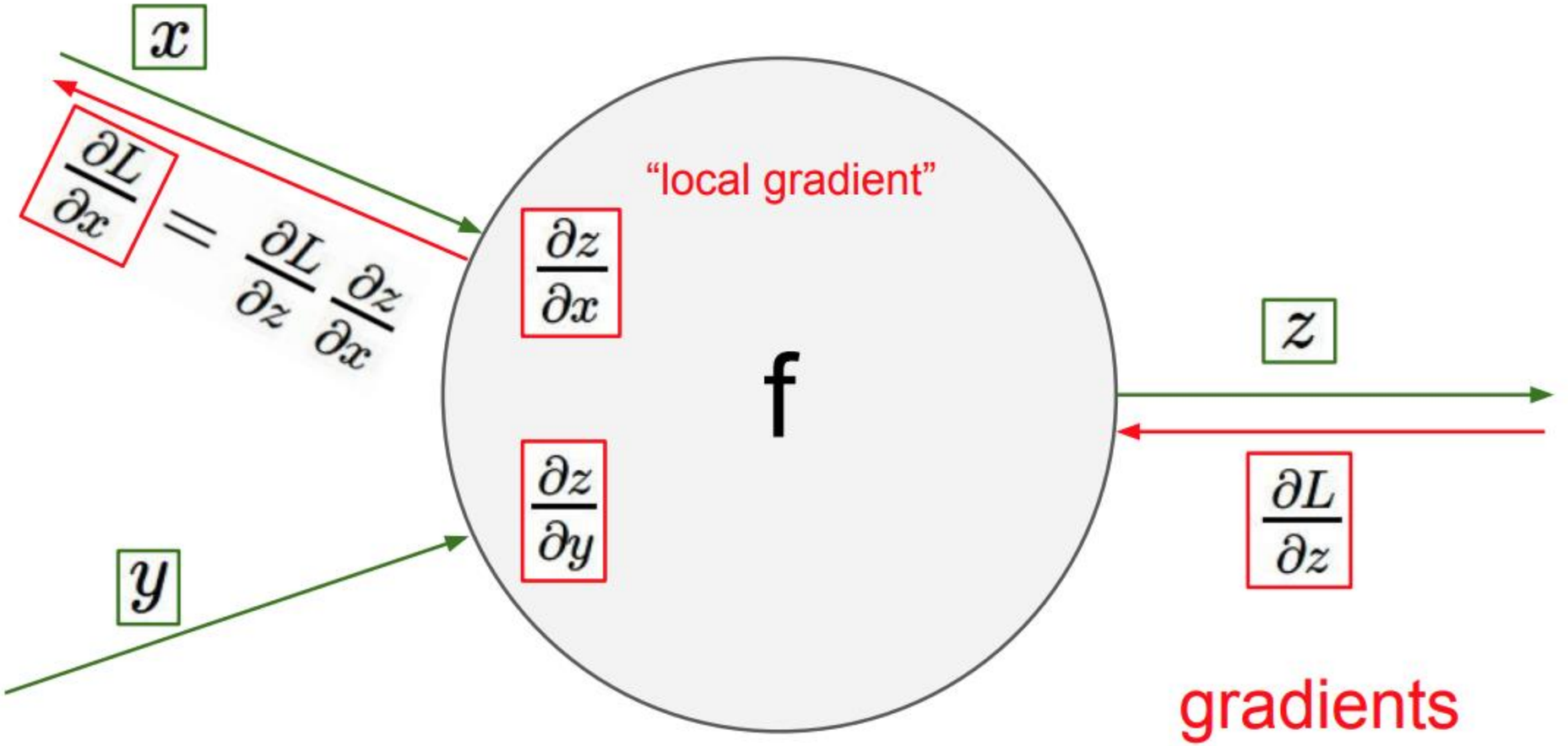
recursion: j is a hidden unit

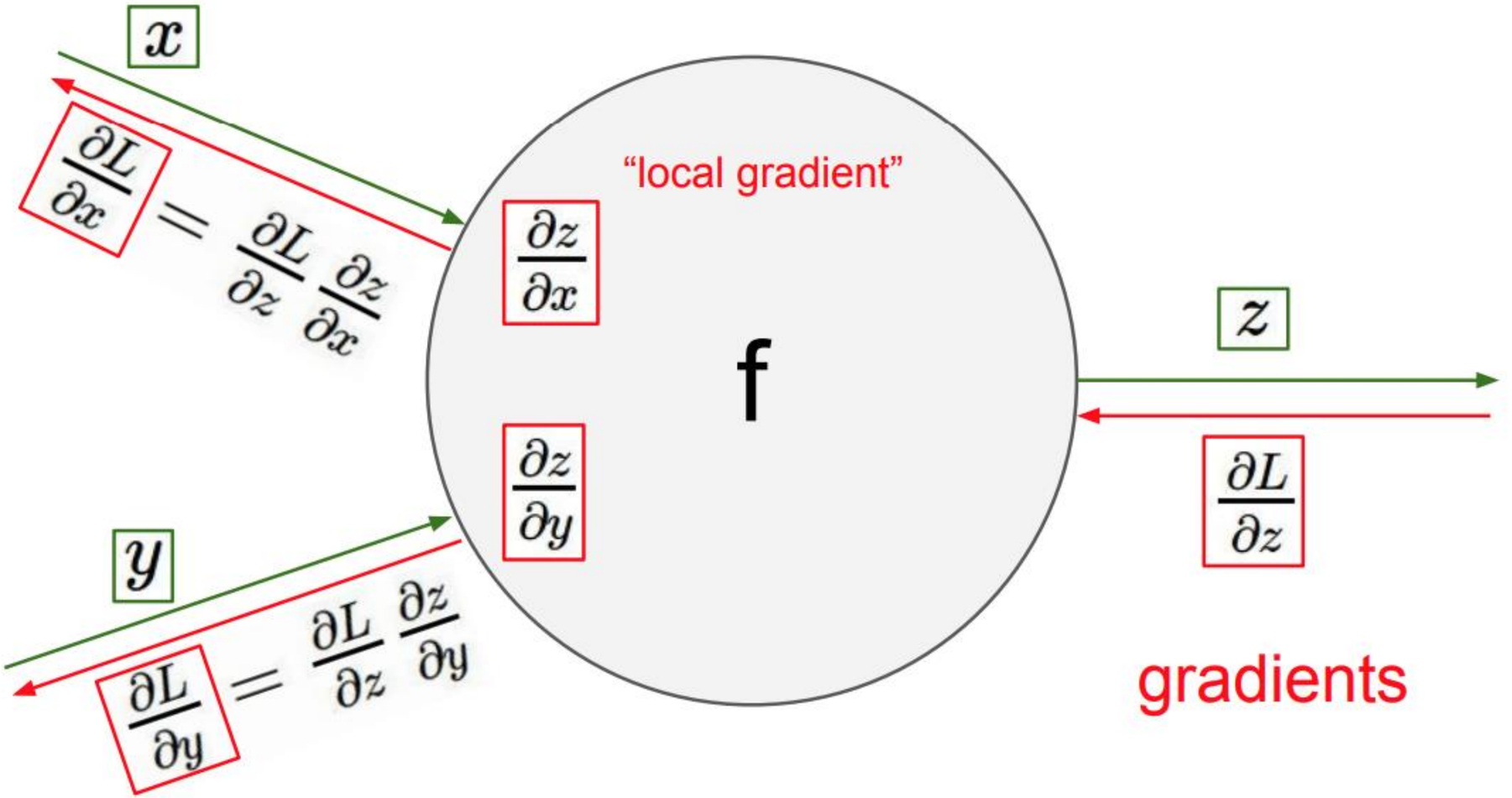
Explanation in 3Blue1Brown about the backpropagation algorithm
https://www.youtube.com/watch?v=Ilg3gGewQ5U


x
y
f
z


x
y
“local gradient”
$\frac{\partial z}{\partial x}$
$\frac{\partial z}{\partial y}$
f
z


x
y
"local gradient"
$\frac{\partial z}{\partial x}$
$\frac{\partial z}{\partial y}$
f
z
$\frac{\partial L}{\partial z}$
gradients


x
$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z}\frac{\partial z}{\partial x}$
"local gradient"
$\frac{\partial z}{\partial x}$
f
$\frac{\partial z}{\partial y}$
y
z
$\frac{\partial L}{\partial z}$
gradients



Deep learning frameworks use automatic differentiation

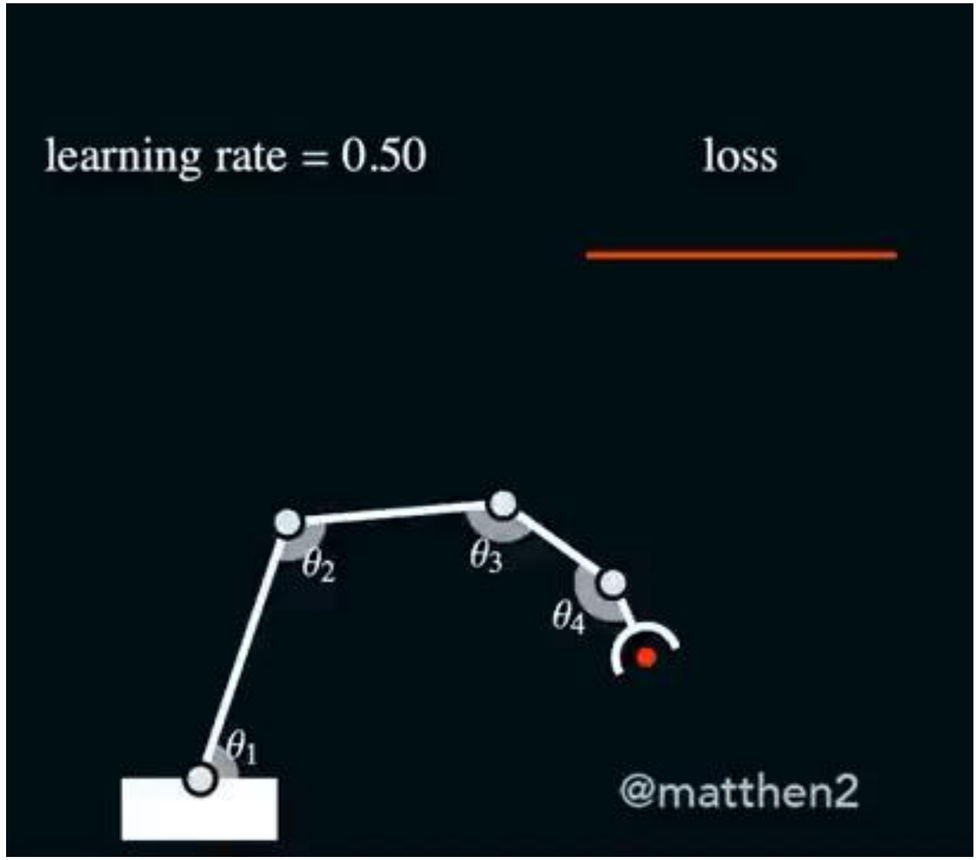
Vanishing and exploding gradients

- Diagnosis:
  - **Exploding**: model not learning; cost/loss oscillating or NaN, weights growing exponentially to NaN
  - **Vanishing**: learning very slow or even stopped very early; weights on the last layers change but those closer to the input don't
- Popular solutions:
  - Limiting the size of the gradients (gradient clipping)
  - Pre-training or proper weight initialization (e.g. Xavier, Kaiming for ReLU layers)
  - Skip connections
  - Batch normalization

Adapted from José Luis Alba

Learning rate

- LR is the most important hyperparameter in backpropagation and the most **difficult** to set
- Affects: speed of learning, stabilization, escaping from local minima, etc.
  - Too small → slow training, difficult to escape from small gradient areas
  - Too large → oscillating too much (or not converging), not reaching narrow minimums
- Learning rate strategies:
  - Initial large LR to adapt quickly to the initial random weights
  - Modify the LR during training (many possible strategies)



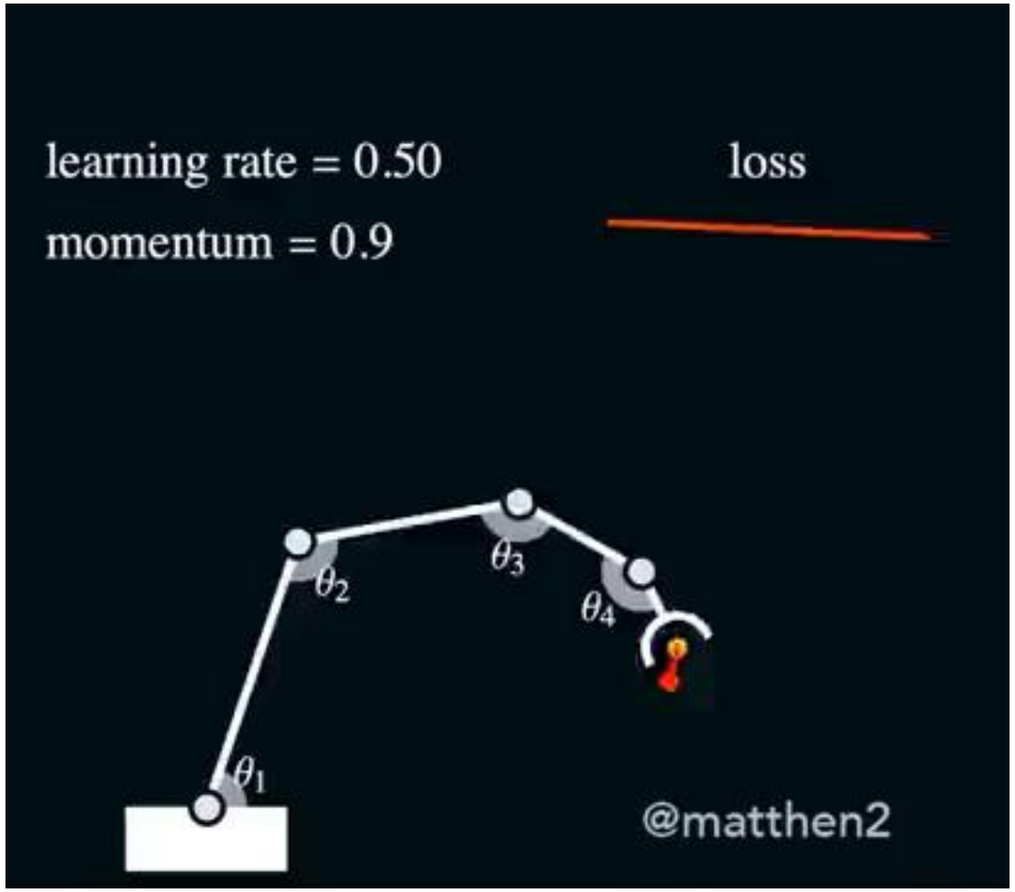
https://twitter.com/matthen2/status/1520427516997025792

Faster convergence

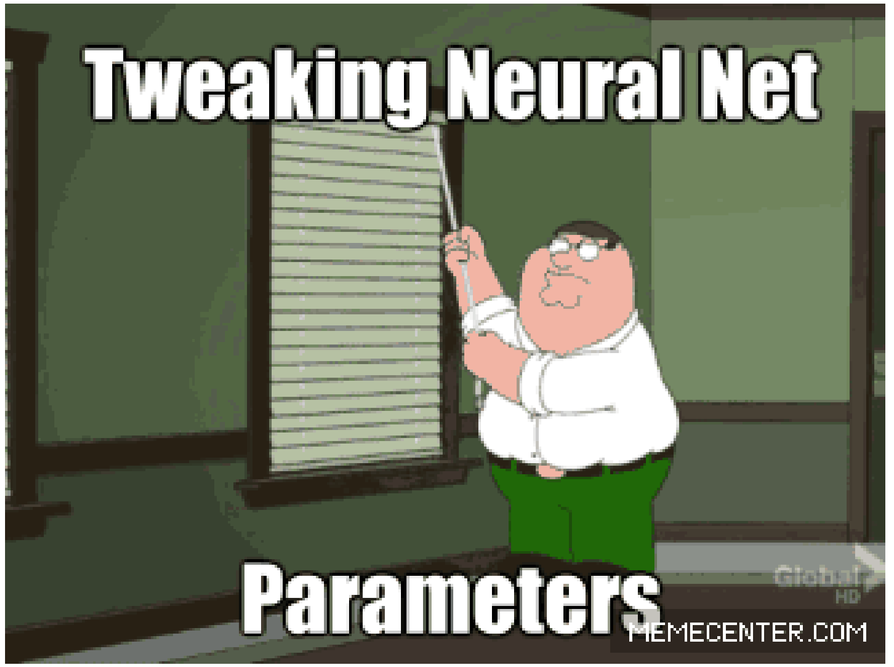
- **Adaptive gradients**
  - Idea: adjust the learning rate of each dimension separately
  - AdaGrad, RMSprop

- **Adaptive moment estimation**
  - Idea: also replace gradient by its moving average to induce momentum
  - Adam

adapted from Pascal Poupart



https://twitter.com/matthen2/status/1520431854180081672


Tweaking Neural Net
Parameters
MEMECENTER.COM

Overfitting

- There is serious risk of **overfitting**, since typically the number of parameters is much larger than the amount of data

- Some solutions:
  - Early stopping
  - Regularization
    - L1 or L2 penalty term in the Loss (penalizes large weights)
    - Dropout
    - Data augmentation

Overfitting

- Dropout Perturbations in the network
  - randomly “drop” some units from the network when training

- Data augmentation Perturbations in the input data
  - create new data by applying transformations to the given dataset

Models

- Unsupervised learning (unlabelled data)
  - Restricted Boltzman Machine (RBM)
  - Autoencoder
- Supervised learning (e.g. classification)
  - Deep Belief Network (DBN)
  - Convolutional Neural Network (CNN)
- Learning in a time series (e.g. forecasting)
  - Recurrent network (e.g. LSTM)